FML - Clustering - April1st

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Loading dataset into R environment

```
pharma <- read.csv("/Users/roc/Downloads/Pharmaceuticals-1.csv")
```

Understanding basics of finacial data

```
str(pharma)
```

```
## 'data.frame':    21 obs. of  14 variables:
##  $ Symbol              : chr  "ABT" "AGN" "AHM" "AZN" ...
##  $ Name                : chr  "Abbott Laboratories" "Allergan, Inc." "Amersham
plc" "AstraZeneca PLC" ...
##  $ Market_Cap          : num  68.44 7.58 6.3 67.63 47.16 ...
##  $ Beta                : num  0.32 0.41 0.46 0.52 0.32 1.11 0.5 0.85 1.08 0.18
...
##  $ PE_Ratio            : num  24.7 82.5 20.7 21.5 20.1 27.9 13.9 26 3.6 27.9
...
##  $ ROE                 : num  26.4 12.9 14.9 27.4 21.8 3.9 34.8 24.1 15.1 31
...
##  $ ROA                 : num  11.8 5.5 7.8 15.4 7.5 1.4 15.1 4.3 5.1 13.5 ...
##  $ Asset_Turnover      : num  0.7 0.9 0.9 0.9 0.6 0.6 0.9 0.6 0.3 0.6 ...
##  $ Leverage            : num  0.42 0.6 0.27 0 0.34 0 0.57 3.51 1.07 0.53 ...
##  $ Rev_Growth          : num  7.54 9.16 7.05 15 26.81 ...
##  $ Net_Profit_Margin   : num  16.1 5.5 11.2 18 12.9 2.6 20.6 7.5 13.3 23.4 ...
##  $ Median_Recommendation: chr  "Moderate Buy" "Moderate Buy" "Strong Buy" "Mode
rate Sell" ...
##  $ Location            : chr  "US" "CANADA" "UK" "UK" ...
##  $ Exchange            : chr  "NYSE" "NYSE" "NYSE" "NYSE" ...
```

```
summary(pharma)
```

```
##     Symbol              Name             Market_Cap          Beta
##  Length:21          Length:21          Min.   :  0.41   Min.   :0.1800
##  Class :character   Class :character   1st Qu.:  6.30   1st Qu.:0.3500
##  Mode  :character   Mode  :character   Median : 48.19   Median :0.4600
##                                        Mean   : 57.65   Mean   :0.5257
##                                        3rd Qu.: 73.84   3rd Qu.:0.6500
##                                        Max.   :199.47   Max.   :1.1100
##     PE_Ratio          ROE            ROA        Asset_Turnover    Leverage
##  Min.   : 3.60   Min.   : 3.9   Min.   : 1.40   Min.   :0.3    Min.   :0.0000
##  1st Qu.:18.90   1st Qu.:14.9   1st Qu.: 5.70   1st Qu.:0.6    1st Qu.:0.1600
##  Median :21.50   Median :22.6   Median :11.20   Median :0.6    Median :0.3400
##  Mean   :25.46   Mean   :25.8   Mean   :10.51   Mean   :0.7    Mean   :0.5857
##  3rd Qu.:27.90   3rd Qu.:31.0   3rd Qu.:15.00   3rd Qu.:0.9    3rd Qu.:0.6000
##  Max.   :82.50   Max.   :62.9   Max.   :20.30   Max.   :1.1    Max.   :3.5100
##    Rev_Growth    Net_Profit_Margin Median_Recommendation   Location
##  Min.   :-3.17   Min.   : 2.6      Length:21             Length:21
##  1st Qu.: 6.38   1st Qu.:11.2      Class :character      Class :character
##  Median : 9.37   Median :16.1      Mode  :character      Mode  :character
##  Mean   :13.37   Mean   :15.7
##  3rd Qu.:21.87   3rd Qu.:21.1
##  Max.   :34.21   Max.   :25.5
##    Exchange
##  Length:21
##  Class :character
##  Mode  :character
##
##
##
```

PART (a):Cluster using only numerical variables

```
df_num <- pharma[, 3:11]
rownames(df_num) <- pharma$Name
print(df_num)
```

```
##                                   Market_Cap Beta PE_Ratio  ROE  ROA
## Abbott Laboratories                    68.44 0.32     24.7 26.4 11.8
## Allergan, Inc.                          7.58 0.41     82.5 12.9  5.5
## Amersham plc                            6.30 0.46     20.7 14.9  7.8
## AstraZeneca PLC                        67.63 0.52     21.5 27.4 15.4
## Aventis                                47.16 0.32     20.1 21.8  7.5
## Bayer AG                               16.90 1.11     27.9  3.9  1.4
## Bristol-Myers Squibb Company           51.33 0.50     13.9 34.8 15.1
## Chattem, Inc                            0.41 0.85     26.0 24.1  4.3
## Elan Corporation, plc                   0.78 1.08      3.6 15.1  5.1
```

```
## Eli Lilly and Company                    73.84 0.18     27.9 31.0 13.5
## GlaxoSmithKline plc                     122.11 0.35     18.0 62.9 20.3
## IVAX Corporation                          2.60 0.65     19.9 21.4  6.8
## Johnson & Johnson                       173.93 0.46     28.4 28.6 16.3
## Medicis Pharmaceutical Corporation        1.20 0.75     28.6 11.2  5.4
## Merck & Co., Inc.                       132.56 0.46     18.9 40.6 15.0
## Novartis AG                              96.65 0.19     21.6 17.9 11.2
## Pfizer Inc                              199.47 0.65     23.6 45.6 19.2
## Pharmacia Corporation                    56.24 0.40     56.5 13.5  5.7
## Schering-Plough Corporation              34.10 0.51     18.9 22.6 13.3
## Watson Pharmaceuticals, Inc.              3.26 0.24     18.4 10.2  6.8
## Wyeth                                    48.19 0.63     13.1 54.9 13.4
##                                   Asset_Turnover Leverage Rev_Growth
## Abbott Laboratories                          0.7     0.42       7.54
## Allergan, Inc.                               0.9     0.60       9.16
## Amersham plc                                 0.9     0.27       7.05
## AstraZeneca PLC                              0.9     0.00      15.00
## Aventis                                      0.6     0.34      26.81
## Bayer AG                                     0.6     0.00      -3.17
## Bristol-Myers Squibb Company                 0.9     0.57       2.70
## Chattem, Inc                                 0.6     3.51       6.38
## Elan Corporation, plc                        0.3     1.07      34.21
## Eli Lilly and Company                        0.6     0.53       6.21
## GlaxoSmithKline plc                          1.0     0.34      21.87
## IVAX Corporation                             0.6     1.45      13.99
## Johnson & Johnson                            0.9     0.10       9.37
## Medicis Pharmaceutical Corporation           0.3     0.93      30.37
## Merck & Co., Inc.                            1.1     0.28      17.35
## Novartis AG                                  0.5     0.06      -2.69
## Pfizer Inc                                   0.8     0.16      25.54
## Pharmacia Corporation                        0.6     0.35      15.00
## Schering-Plough Corporation                  0.8     0.00       8.56
## Watson Pharmaceuticals, Inc.                 0.5     0.20      29.18
## Wyeth                                        0.6     1.12       0.36
##                                   Net_Profit_Margin
## Abbott Laboratories                            16.1
## Allergan, Inc.                                  5.5
## Amersham plc                                   11.2
## AstraZeneca PLC                                18.0
## Aventis                                        12.9
## Bayer AG                                        2.6
## Bristol-Myers Squibb Company                   20.6
## Chattem, Inc                                    7.5
## Elan Corporation, plc                          13.3
## Eli Lilly and Company                          23.4
## GlaxoSmithKline plc                            21.1
## IVAX Corporation                               11.0
## Johnson & Johnson                              17.9
## Medicis Pharmaceutical Corporation             21.3
## Merck & Co., Inc.                              14.1
```

```
## Novartis AG                                        22.4
## Pfizer Inc                                         25.2
## Pharmacia Corporation                               7.3
## Schering-Plough Corporation                        17.6
## Watson Pharmaceuticals, Inc.                       15.1
## Wyeth                                              25.5
```

Z-Score Normalization

Now I scale the data using z-score normalization to remove scale bias. Without scaling, variables with larger ranges would dominate the distance calculations unfairly.

```
df_scaled <- scale(df_num)
print(df_scaled)
```

```
##                                    Market_Cap        Beta    PE_Ratio
## Abbott Laboratories                 0.1840960 -0.80125356 -0.04671323
## Allergan, Inc.                     -0.8544181 -0.45070513  3.49706911
## Amersham plc                       -0.8762600 -0.25595600 -0.29195768
## AstraZeneca PLC                     0.1702742 -0.02225704 -0.24290879
## Aventis                            -0.1790256 -0.80125356 -0.32874435
## Bayer AG                           -0.6953818  2.27578267  0.14948233
## Bristol-Myers Squibb Company       -0.1078688 -0.10015669 -0.70887325
## Chattem, Inc                       -0.9767669  1.26308721  0.03299122
## Elan Corporation, plc              -0.9704532  2.15893320 -1.34037772
## Eli Lilly and Company               0.2762415 -1.34655112  0.14948233
## GlaxoSmithKline plc                 1.0999201 -0.68440408 -0.45749769
## IVAX Corporation                   -0.9393967  0.48409069 -0.34100657
## Johnson & Johnson                   1.9841758 -0.25595600  0.18013789
## Medicis Pharmaceutical Corporation -0.9632863  0.87358895  0.19240011
## Merck & Co., Inc.                   1.2782387 -0.25595600 -0.40231769
## Novartis AG                         0.6654710 -1.30760129 -0.23677768
## Pfizer Inc                          2.4199899  0.48409069 -0.11415545
## Pharmacia Corporation              -0.0240846 -0.48965495  1.90298017
## Schering-Plough Corporation        -0.4018812 -0.06120687 -0.40231769
## Watson Pharmaceuticals, Inc.       -0.9281345 -1.11285216 -0.43297324
## Wyeth                              -0.1614497  0.40619104 -0.75792214
##                                            ROE        ROA Asset_Turnover
## Abbott Laboratories                 0.04009035  0.2416121  -5.121077e-16
```

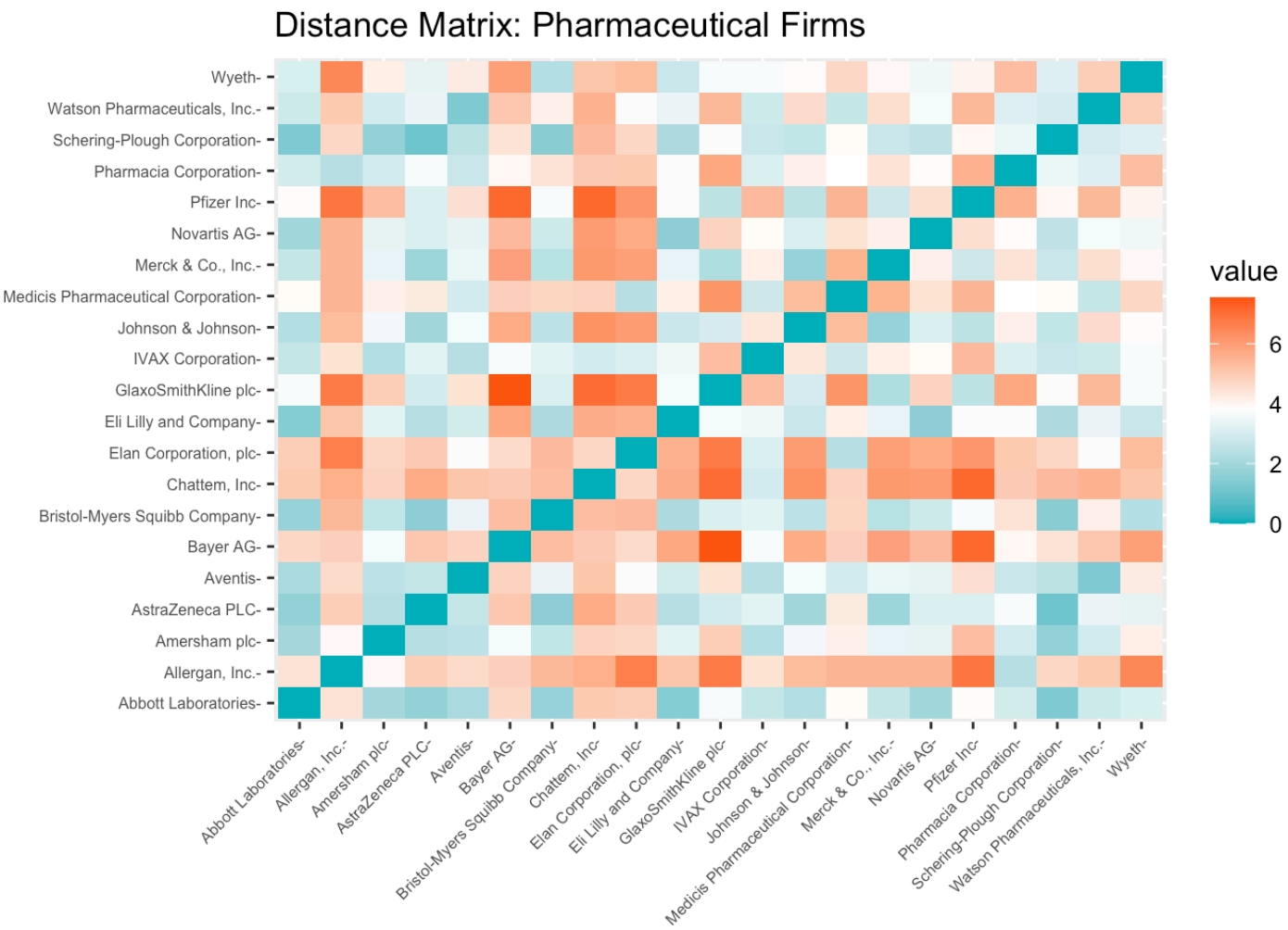
```
## Allergan, Inc.                      -0.85483986 -0.9422871   9.225312e-01
## Amersham plc                        -0.72225761 -0.5100700   9.225312e-01
## AstraZeneca PLC                      0.10638147  0.9181259   9.225312e-01
## Aventis                             -0.26484883 -0.5664461  -4.612656e-01
## Bayer AG                            -1.45146000 -1.7127612  -4.612656e-01
## Bristol-Myers Squibb Company         0.59693581  0.8617498   9.225312e-01
## Chattem, Inc                        -0.11237924 -1.1677918  -4.612656e-01
## Elan Corporation, plc               -0.70899938 -1.0174553  -1.845062e+00
## Eli Lilly and Company                0.34502953  0.5610770  -4.612656e-01
## GlaxoSmithKline plc                  2.45971647  1.8389364   1.383797e+00
## IVAX Corporation                    -0.29136529 -0.6979905  -4.612656e-01
## Johnson & Johnson                    0.18593083  1.0872544   9.225312e-01
## Medicis Pharmaceutical Corporation -0.96753478 -0.9610792  -1.845062e+00
## Merck & Co., Inc.                    0.98142435  0.8429577   1.845062e+00
## Novartis AG                         -0.52338423  0.1288598  -9.225312e-01
## Pfizer Inc                           1.31287998  1.6322239   4.612656e-01
## Pharmacia Corporation               -0.81506519 -0.9047030  -4.612656e-01
## Schering-Plough Corporation         -0.21181593  0.5234929   4.612656e-01
## Watson Pharmaceuticals, Inc.        -1.03382590 -0.6979905  -9.225312e-01
## Wyeth                                1.92938746  0.5422849  -4.612656e-01
##                                        Leverage  Rev_Growth Net_Profit_Margin
## Abbott Laboratories                 -0.21209793 -0.52776752        0.06168225
## Allergan, Inc.                       0.01828430 -0.38113909       -1.55366706
## Amersham plc                        -0.40408312 -0.57211809       -0.68503583
## AstraZeneca PLC                     -0.74965647  0.14744734        0.35122600
## Aventis                             -0.31449003  1.21638667       -0.42597037
## Bayer AG                            -0.74965647 -1.49714434       -1.99560225
## Bristol-Myers Squibb Company        -0.02011273 -0.96584257        0.74744375
## Chattem, Inc                         3.74279705 -0.63276071       -1.24888417
## Elan Corporation, plc                0.61983791  1.88617085       -0.36501379
## Eli Lilly and Company               -0.07130879 -0.64814764        1.17413980
## GlaxoSmithKline plc                 -0.31449003  0.76926048        0.82363947
## IVAX Corporation                     1.10620040  0.05603085       -0.71551412
## Johnson & Johnson                   -0.62166634 -0.36213170        0.33598685
## Medicis Pharmaceutical Corporation   0.44065173  1.53860717        0.85411776
## Merck & Co., Inc.                   -0.39128411  0.36014907       -0.24310064
## Novartis AG                         -0.67286239 -1.45369888        1.02174835
## Pfizer Inc                          -0.54487226  1.10143723        1.44844440
## Pharmacia Corporation               -0.30169102  0.14744734       -1.27936246
## Schering-Plough Corporation         -0.74965647 -0.43544591        0.29026942
## Watson Pharmaceuticals, Inc.        -0.49367621  1.43089863       -0.09070919
## Wyeth                                0.68383297 -1.17763919        1.49416183
## attr(,"scaled:center")
##        Market_Cap              Beta          PE_Ratio               ROE
##        57.6514286         0.5257143        25.4619048        25.7952381
##               ROA    Asset_Turnover          Leverage        Rev_Growth
##        10.5142857         0.7000000         0.5857143        13.3709524
## Net_Profit_Margin
##        15.6952381
## attr(,"scaled:scale")
```

```
##        Market_Cap              Beta          PE_Ratio               ROE
##        58.6029595         0.2567406        16.3102568        15.0849752
##               ROA    Asset_Turnover          Leverage        Rev_Growth
##         5.3213988         0.2167948         0.7813103        11.0483351
## Net_Profit_Margin
##         6.5620482
```

Now I compute Euclidean distances between all pairs of firms

```
distance <- get_dist(df_scaled)
```

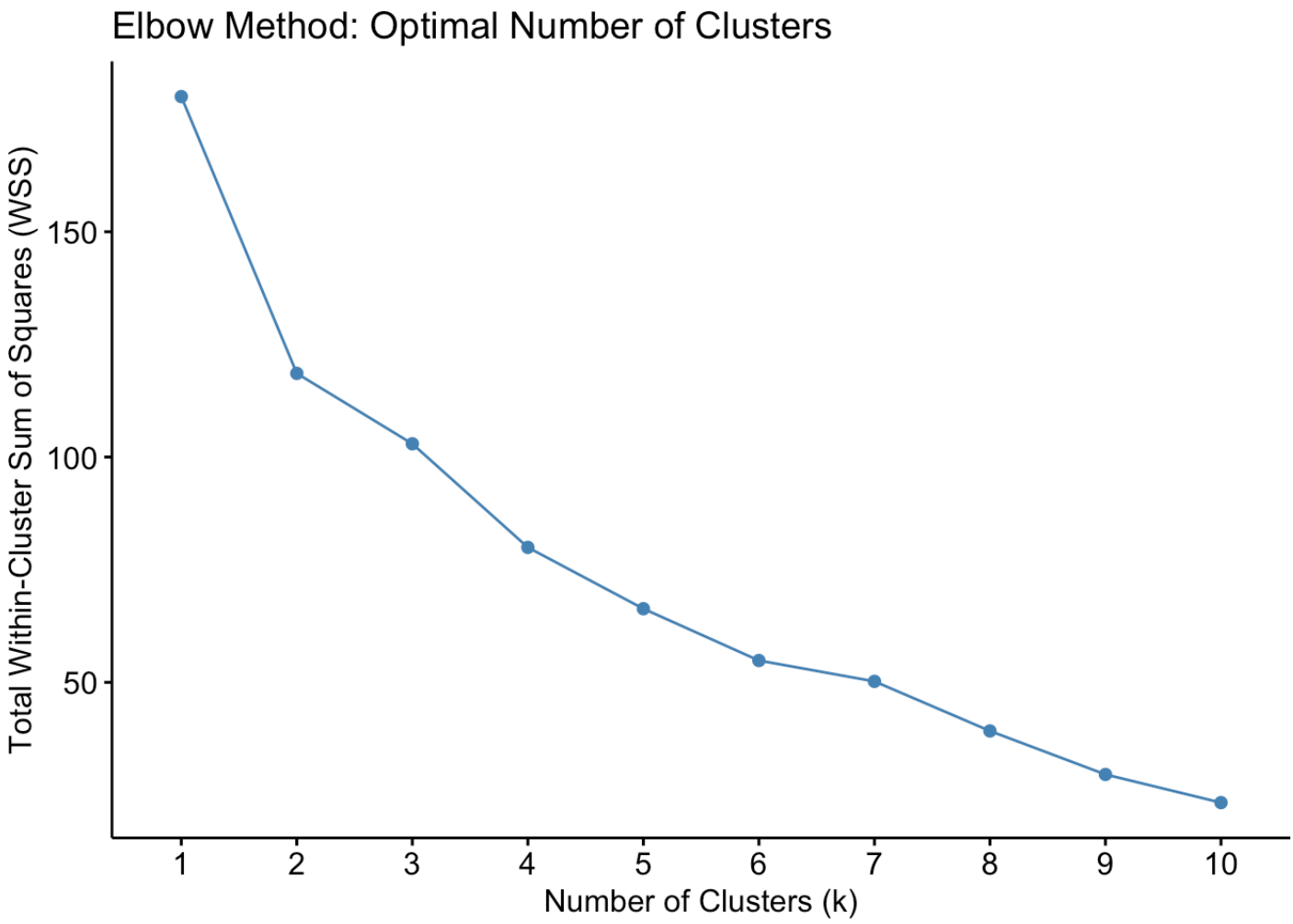
```
fviz_dist(distance,gradient= list(low="#00AFBB",mid = "white", high= "#FC4E07"),
          lab_size = 6) +labs(title ="Distance Matrix: Pharmaceutical Firms")
```



Now I determine Optimal Number of Clusters (k)

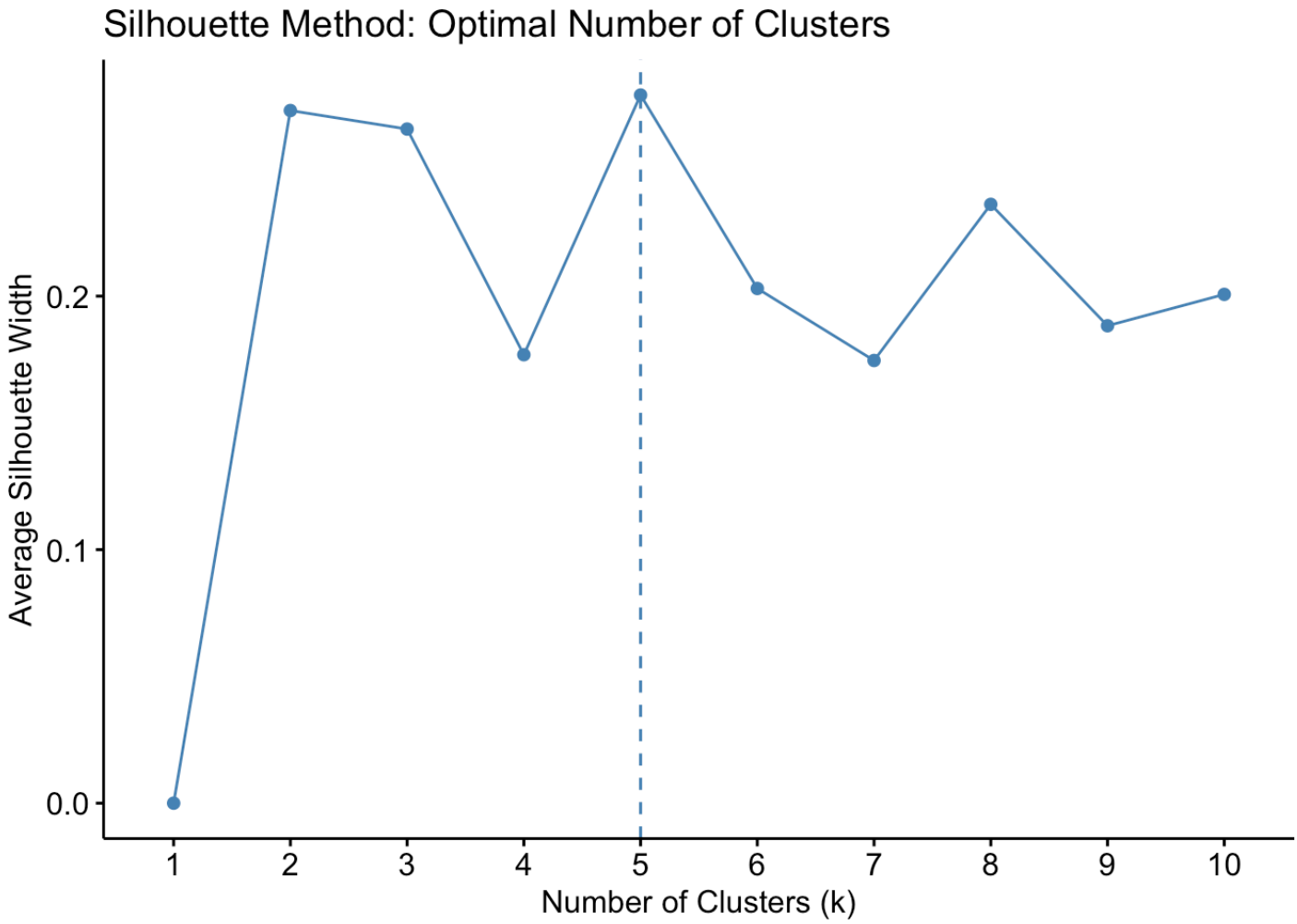
Method-1: Elbow Method

```
set.seed(123)
fviz_nbclust(df_scaled,kmeans, method = "wss") +labs(title= "Elbow Method: Optimal
Number of Clusters",x ="Number of Clusters (k)",y = "Total Within-Cluster Sum of S
quares (WSS)")
```



Method 2: Silhouette Method

```
set.seed(123)
fviz_nbclust(df_scaled,kmeans, method = "silhouette") +labs(title ="Silhouette Met
hod: Optimal Number of Clusters",x ="Number of Clusters (k)",y ="Average Silhouett
e Width")
```



Now I run K-Means clustering with k=5. Based on Elbow and Silhouette methods, k = 5 is selected as optimal.

```
set.seed(123)
k5 <- kmeans(df_scaled,centers =5,nstart= 25)
print(k5$centers)
```

```
##    Market_Cap       Beta    PE_Ratio        ROE        ROA Asset_Turnover
## 1 -0.03142211 -0.4360989 -0.31724852  0.1950459  0.4083915      0.1729746
## 2 -0.87051511  1.3409869 -0.05284434 -0.6184015 -1.1928478     -0.4612656
## 3 -0.43925134 -0.4701800  2.70002464 -0.8349525 -0.9234951      0.2306328
## 4  1.69558112 -0.1780563 -0.19845823  1.2349879  1.3503431      1.1531640
## 5 -0.76022489  0.2796041 -0.47742380 -0.7438022 -0.8107428     -1.2684804
##      Leverage Rev_Growth Net_Profit_Margin
## 1 -0.27449312 -0.7041516       0.556954446
## 2  1.36644699 -0.6912914      -1.320000179
## 3 -0.14170336 -0.1168459      -1.416514761
## 4 -0.46807818  0.4671788       0.591242521
## 5  0.06308085  1.5180158      -0.006893899
```

to view size of each cluster

```
print(k5$size)
```

```
## [1] 8 3 2 4 4
```

to view cluster assignment for each firm

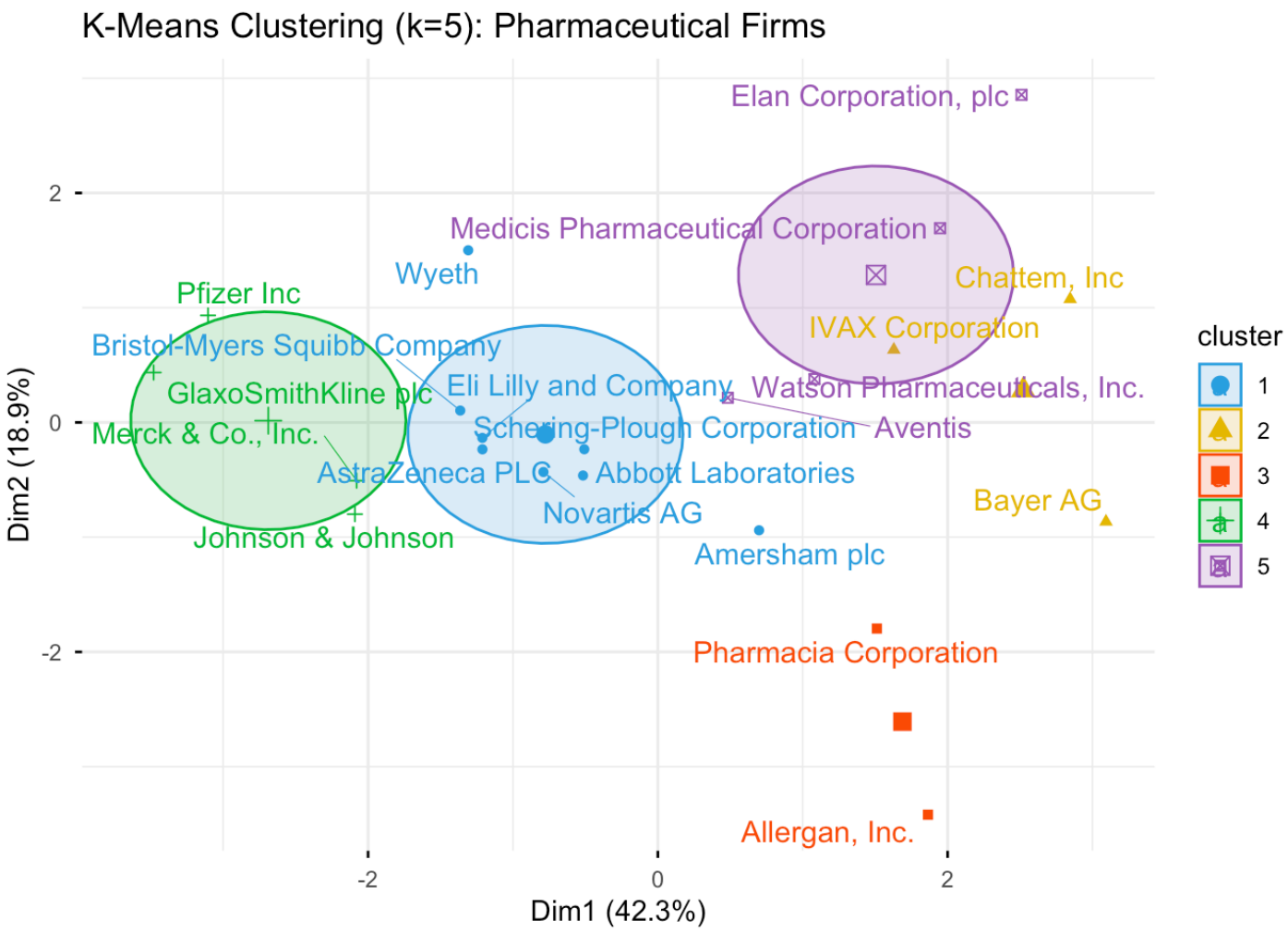
```
print(k5$cluster)
```

```
##                Abbott Laboratories                      Allergan, Inc.
##                                  1                                   3
##                      Amersham plc                     AstraZeneca PLC
##                                  1                                   1
##                           Aventis                           Bayer AG
##                                  5                                   2
##       Bristol-Myers Squibb Company                       Chattem, Inc
##                                  1                                   2
##              Elan Corporation, plc              Eli Lilly and Company
##                                  5                                   1
##                GlaxoSmithKline plc                   IVAX Corporation
##                                  4                                   2
##                  Johnson & Johnson Medicis Pharmaceutical Corporation
##                                  4                                   5
##                  Merck & Co., Inc.                        Novartis AG
##                                  4                                   1
##                        Pfizer Inc              Pharmacia Corporation
##                                  4                                   3
##        Schering-Plough Corporation       Watson Pharmaceuticals, Inc.
##                                  1                                   5
##                             Wyeth
##                                  1
```

Now I visualize the clusters

```
fviz_cluster(k5, data=df_scaled,palette = c("#2E9FDF","#E7B800","#FC4E07","#00BA3
8","#9b59b6"), ellipse.type= "euclid",repel=TRUE,ggtheme = theme_minimal()) +
  labs(title ="K-Means Clustering (k=5): Pharmaceutical Firms")
```

```
## Too few points to calculate an ellipse
## Too few points to calculate an ellipse
```



PART (b): Interpretetion of clusters based on numerical variables

```
pharma$Cluster <- k5$cluster
cluster_summary <- pharma %>%
  group_by(Cluster) %>%
  summarise(
    Firms        = n(),
    Avg_MarketCap =mean(Market_Cap),
    Avg_Beta      =mean(Beta),
    Avg_PE_Ratio  =mean(PE_Ratio),
    Avg_ROE       =mean(ROE),
    Avg_ROA       =mean(ROA),
    Avg_AssetTO   =mean(Asset_Turnover),
    Avg_Leverage  =mean(Leverage),
    Avg_RevGrowth =mean(Rev_Growth),
    Avg_NetMargin =mean(Net_Profit_Margin))
print(cluster_summary)
```

```
## # A tibble: 5 × 11
##   Cluster Firms Avg_MarketCap Avg_Beta Avg_PE_Ratio Avg_ROE Avg_ROA Avg_AssetTO
##     <int> <int>         <dbl>    <dbl>        <dbl>   <dbl>   <dbl>       <dbl>
## 1       1     8          55.8     0.414         20.3    28.7   12.7        0.738
## 2       2     3           6.64    0.87          24.6    16.5    4.17       0.6
## 3       3     2          31.9     0.405         69.5    13.2    5.6        0.75
## 4       4     4         157.      0.48          22.2    44.4   17.7        0.95
## 5       5     4          13.1     0.598         17.7    14.6    6.2        0.425
## # ℹ 3 more variables: Avg_Leverage <dbl>, Avg_RevGrowth <dbl>,
## #   Avg_NetMargin <dbl>
```

```
pharma %>%
  select(Name, Cluster) %>%
  arrange(Cluster) %>%
  print()
```

```
##                                 Name Cluster
## 1                 Abbott Laboratories       1
## 2                        Amersham plc       1
## 3                     AstraZeneca PLC       1
## 4        Bristol-Myers Squibb Company       1
## 5               Eli Lilly and Company       1
## 6                          Novartis AG       1
## 7         Schering-Plough Corporation       1
## 8                               Wyeth       1
## 9                            Bayer AG       2
## 10                       Chattem, Inc       2
## 11                    IVAX Corporation       2
## 12                      Allergan, Inc.       3
## 13              Pharmacia Corporation       3
## 14                GlaxoSmithKline plc       4
## 15                  Johnson & Johnson       4
## 16                  Merck & Co., Inc.       4
## 17                         Pfizer Inc       4
## 18                            Aventis       5
## 19              Elan Corporation, plc       5
## 20 Medicis Pharmaceutical Corporation       5
## 21       Watson Pharmaceuticals, Inc.       5
```

PART (c): Pattern in non-numerical variables(10-12)

```
cat("\n Median Recommendation by Cluster \n")
```

```
##
##  Median Recommendation by Cluster
```

```
pharma %>%
  select(Name, Cluster, Median_Recommendation) %>%
  arrange(Cluster) %>%
  print()
```

```
##                                 Name Cluster Median_Recommendation
## 1                 Abbott Laboratories       1          Moderate Buy
## 2                        Amersham plc       1            Strong Buy
## 3                     AstraZeneca PLC       1         Moderate Sell
## 4        Bristol-Myers Squibb Company       1         Moderate Sell
## 5               Eli Lilly and Company       1                  Hold
## 6                         Novartis AG       1                  Hold
## 7         Schering-Plough Corporation       1                  Hold
## 8                               Wyeth       1                  Hold
## 9                            Bayer AG       2                  Hold
## 10                       Chattem, Inc       2          Moderate Buy
## 11                    IVAX Corporation       2                  Hold
## 12                      Allergan, Inc.       3          Moderate Buy
## 13              Pharmacia Corporation       3                  Hold
## 14                GlaxoSmithKline plc       4                  Hold
## 15                  Johnson & Johnson       4          Moderate Buy
## 16                  Merck & Co., Inc.       4                  Hold
## 17                         Pfizer Inc       4          Moderate Buy
## 18                            Aventis       5          Moderate Buy
## 19              Elan Corporation, plc       5         Moderate Sell
## 20 Medicis Pharmaceutical Corporation       5          Moderate Buy
## 21       Watson Pharmaceuticals, Inc.       5         Moderate Sell
```

Now I count each recommendation type per cluster

```
rec_pattern <- pharma %>%
  group_by(Cluster, Median_Recommendation) %>%
  summarise(Count = n(), .groups = "drop")
print(rec_pattern)
```

```
## # A tibble: 12 × 3
##    Cluster Median_Recommendation Count
##      <int> <chr>                 <int>
##  1       1 Hold                      4
##  2       1 Moderate Buy              1
##  3       1 Moderate Sell             2
##  4       1 Strong Buy                1
##  5       2 Hold                      2
##  6       2 Moderate Buy              1
##  7       3 Hold                      1
##  8       3 Moderate Buy              1
##  9       4 Hold                      2
## 10       4 Moderate Buy              2
## 11       5 Moderate Buy              2
## 12       5 Moderate Sell             2
```

Pattern: Location (HQ Country) per Cluster

```
cat("\n Location (HQ Country) by Cluster \n")
```

```
##
##  Location (HQ Country) by Cluster
```

```
loc_pattern <- pharma %>%
  group_by(Cluster, Location) %>%
  summarise(Count = n(), .groups = "drop")
print(loc_pattern)
```

```
## # A tibble: 12 × 3
##    Cluster Location    Count
##      <int> <chr>       <int>
##  1       1 SWITZERLAND     1
##  2       1 UK              2
##  3       1 US              5
##  4       2 GERMANY         1
##  5       2 US              2
##  6       3 CANADA          1
##  7       3 US              1
##  8       4 UK              1
##  9       4 US              3
## 10       5 FRANCE          1
## 11       5 IRELAND         1
## 12       5 US              2
```

Pattern: Stock Exchange per Cluster

```
cat("\n Stock Exchange by Cluster \n")
```

```
##
##  Stock Exchange by Cluster
```

```
exc_pattern <- pharma %>%
  group_by(Cluster, Exchange) %>%
  summarise(Count = n(), .groups = "drop")
print(exc_pattern)
```

```
## # A tibble: 7 × 3
##   Cluster Exchange Count
##     <int> <chr>    <int>
## 1       1 NYSE         8
## 2       2 AMEX         1
## 3       2 NASDAQ       1
## 4       2 NYSE         1
## 5       3 NYSE         2
## 6       4 NYSE         4
## 7       5 NYSE         4
```

PART (d): Name Each Cluster

Based on cluster_summary above, I assign descriptive names.These names reflect dominant financial characteristics of each group.

```
cluster_names <- data.frame(
  Cluster = 1:5,
  Cluster_Name = c(
    "Large-Cap Market Leaders",
    "High-Risk Speculative Firms",
    "Undervalued Growth Firms",
    "Profitable Dividend Plays",
    "Small Emerging Challengers"))
```

Merge cluster names into the main dataset

```
pharma_final <- pharma %>%
  left_join(cluster_names, by = "Cluster")
```

Final output: each firm with its cluster number and descriptive name

```
pharma_final %>%
  select(Name, Cluster, Cluster_Name, Median_Recommendation, Location, Exchange) %
>%
  arrange(Cluster) %>%
  print()
```

```
##                                 Name Cluster                Cluster_Name
## 1                 Abbott Laboratories       1    Large-Cap Market Leaders
## 2                        Amersham plc       1    Large-Cap Market Leaders
## 3                     AstraZeneca PLC       1    Large-Cap Market Leaders
## 4        Bristol-Myers Squibb Company       1    Large-Cap Market Leaders
## 5               Eli Lilly and Company       1    Large-Cap Market Leaders
## 6                         Novartis AG       1    Large-Cap Market Leaders
## 7         Schering-Plough Corporation       1    Large-Cap Market Leaders
## 8                               Wyeth       1    Large-Cap Market Leaders
## 9                           Bayer AG       2 High-Risk Speculative Firms
## 10                      Chattem, Inc       2 High-Risk Speculative Firms
## 11                   IVAX Corporation       2 High-Risk Speculative Firms
## 12                     Allergan, Inc.       3    Undervalued Growth Firms
## 13              Pharmacia Corporation       3    Undervalued Growth Firms
## 14                GlaxoSmithKline plc       4   Profitable Dividend Plays
## 15                  Johnson & Johnson       4   Profitable Dividend Plays
## 16                  Merck & Co., Inc.       4   Profitable Dividend Plays
## 17                         Pfizer Inc       4   Profitable Dividend Plays
## 18                            Aventis       5  Small Emerging Challengers
## 19              Elan Corporation, plc       5  Small Emerging Challengers
## 20 Medicis Pharmaceutical Corporation       5  Small Emerging Challengers
## 21       Watson Pharmaceuticals, Inc.       5  Small Emerging Challengers
##    Median_Recommendation    Location Exchange
## 1           Moderate Buy          US     NYSE
## 2             Strong Buy          UK     NYSE
## 3          Moderate Sell          UK     NYSE
## 4          Moderate Sell          US     NYSE
## 5                   Hold          US     NYSE
## 6                   Hold SWITZERLAND     NYSE
## 7                   Hold          US     NYSE
## 8                   Hold          US     NYSE
## 9                   Hold     GERMANY     NYSE
## 10          Moderate Buy          US   NASDAQ
## 11                  Hold          US     AMEX
## 12          Moderate Buy      CANADA     NYSE
## 13                  Hold          US     NYSE
## 14                  Hold          UK     NYSE
## 15          Moderate Buy          US     NYSE
## 16                  Hold          US     NYSE
## 17          Moderate Buy          US     NYSE
## 18          Moderate Buy      FRANCE     NYSE
## 19          Moderate Sell     IRELAND     NYSE
## 20          Moderate Buy          US     NYSE
## 21          Moderate Sell          US     NYSE
```

Brief Summary

The clustering analysis identified five financially distinct categories within pharmaceutical sector. Large-cap companies i.e. Pfizer, J&J, Merck exhibited superior ROE, ROA, and net profit margins, indicating high operational efficiency.

High leverage companies faced increased financial risk due to lower asset turnover. Firms focused on growth demonstrated robust revenue increases but had thinner profit margins, which suggests aggressive reinvestment strategies.Stable mid cap companies maintained a balance of moderate profitability and low beta, implying reduced market risk. Lastly, small-cap firms displayed elevated PE ratios despite yielding modest returns, reflecting market optimism.

In short, firm size and profitability emerged as most significant financial differentiators among five identified clusters.

Five clusters exhibited unique financial characteristics. Companies such as Pfizer,Johnson & Johnson, and Merck were categorized together as large cap, high profitability leaders. Second cluster included high risk, high leverage firms with fluctuating returns. Third cluster represented undervalued firms that demonstrated robust revenue growth but had moderate profit margins. The fourth group comprised stable, high-margin firms that maintained consistent returns on equity. Fifth cluster encompassed small emerging firms with significant growth potential but lower current profitability.

In terms of non-numerical variables, NYSE-listed US firms were prevalent in most clusters, while “Hold” and “Moderate Buy” were the most common broker recommendations across groups suggesting general market confidence in sector.